

MA 5104: Statistical Inference and Learning

Module 1: Statistical Models and Likelihood

Lecture 2: Representing Uncertainty with Probability Models (Lecture 2: Probability: The Language of Uncertainty)

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Key Takeaways

By the end of this lecture, you should be able to

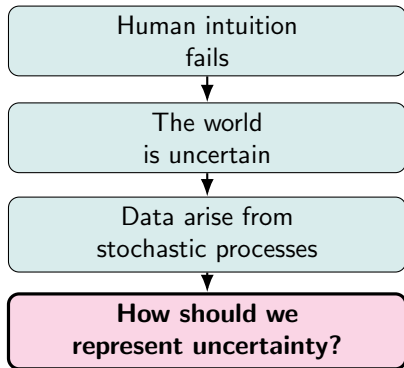
Conceptual Understanding

- Explain why probability is the mathematical language of uncertainty.
- Interpret random variables as mathematical representations of uncertain quantities.
- Distinguish between outcomes, random variables and probability distributions.

Statistical Thinking

- Explain the roles of expectation and variance as summaries of uncertainty.
- Understand joint, marginal and conditional distributions conceptually.
- Recognize that choosing a probability model is a modelling decision based on assumptions.

Where We Left Off



Lecture 1

We established that

- Human intuition is often unreliable.
- Scientific observations exhibit variability.
- Data are generated by stochastic processes.
- Statistical reasoning begins with uncertainty.

Today's Question

If the world is uncertain, how can we represent uncertainty mathematically so that we can reason, predict, and eventually learn from data?

Probability: A Language for Uncertainty

Observation

In many scientific problems, the quantity of interest is **unknown before it is observed**.

Tomorrow's rainfall	Machine lifetime	Customer purchase
?	?	?

Question

How should we mathematically describe quantities whose values are currently unknown?

Answer

Probability provides a principled mathematical framework for representing uncertainty.

Example 1: The Birthday Paradox

Question

Suppose there are **23 people** in a classroom.

What is the probability that at least two people share the same birthday?

- A. Less than 10%
- B. Around 25%
- C. Around 50%
- D. More than 90%

Take a Vote

Do not calculate. Use your intuition.

Birthday Paradox: The Surprise

Answer

For only **23 people**,

$$P(\text{at least one shared birthday}) \approx 0.507$$

More than a 50% chance.

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Why Were We Wrong?

Most people think about **My birthday matching someone else's**.
Instead, we should think about **Every possible pair of people**.

With 23 people,

$$\binom{23}{2} = 253$$

possible pairs exist.

Example 2: Are All Outcomes Equally Likely?

A family has two children. You are told that

At least one child is a boy.

What is the probability that both children are boys?

A. $\frac{1}{2}$

B. $\frac{1}{3}$

C. $\frac{1}{4}$

Take a Vote

Do not compute. Use your intuition.

Equiprobability Bias

The possible outcomes are

$BB, BG, GB, GG.$

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Since at least one child is a boy, the outcome GG is impossible.

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The remaining possibilities are

$$BB, BG, GB,$$

all equally likely. Therefore,

$$P(BB \mid \text{at least one boy}) = \frac{1}{3}.$$

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$$P(BB \mid \text{at least one boy}) = \frac{1}{3}.$$

Most people answer $\frac{1}{2}$.

Why?

Why Did Our Intuition Fail?

Equiprobability Bias

When we know very little, our minds often assume

Everything is equally likely.

even when there is no justification.

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How Statisticians Think

Statisticians do not begin with “Everything is equally likely.” Instead, they ask

What probability model is appropriate?

Key Message

Probability is not guessing. It is a mathematical framework built upon explicit assumptions.

Random Experiments and Sample Spaces

Random Experiment: A **random experiment** is a process whose outcome cannot be predicted with certainty before it is performed, even though the set of all possible outcomes is known.

Sample Space: The **sample space**, denoted by Ω , is the set of all possible outcomes of a random experiment.

Examples

- Toss a coin: $\Omega = \{H, T\}$.
- Roll a die: $\Omega = \{1, 2, 3, 4, 5, 6\}$.
- Lifetime of a battery: $\Omega = [0, \infty)$.

Intuition

The sample space specifies *what could possibly happen*; probability specifies *how likely each possibility is*.

Events

Event: An **event** is any subset of the sample space, $A \subseteq \Omega$.

The event occurs whenever the observed outcome belongs to A .

Example

Rolling a die, $\Omega = \{1, 2, 3, 4, 5, 6\}$

Event: $A = \{\text{Even number}\} = \{2, 4, 6\}$.

Intuition

Probability is assigned to events, not to individual observations. $P(A)$ measures how likely the event is to occur.

Random Variables

Random Variable: A **random variable** is a function $X : \Omega \rightarrow \mathbb{R}$ that assigns a real number to every outcome in the sample space.

Example

Two coin tosses: $\Omega = \{HH, HT, TH, TT\}$.

Let $X =$ Number of heads. Then

$$X(HH) = 2, \quad X(HT) = 1, \quad X(TH) = 1, \quad X(TT) = 0.$$

A random variable is

- neither an event, nor a probability,
- not a random number,
- a mathematical function defined on the sample space.

The randomness comes from the experiment, not from the function itself.

Random Variables

Discrete Random Variables

A random variable is called **discrete** if it assumes only a finite or countably infinite number of values.

Examples: Number of heads, Number of customers, Number of defects.

Continuous Random Variables

A random variable is called **continuous** if it can assume any value in an interval.

Examples: Height, Weight, Time , Temperature.

Discrete	Continuous
Count	Measure
Finite/Countable	Interval
PMF	PDF

Probability Distributions

The **probability distribution** of a random variable completely describes its uncertainty.

For a discrete random variable,

$$p(x) = P(X = x), \quad \text{s.t.} \quad p(x) \geq 0, \quad \sum_{x \in \mathcal{D}} p(x) = 1$$

For a continuous random variable,

$$f(x) \geq 0, \quad \int_{x \in \mathcal{D}} f(x) dx = 1 \quad \text{s.t.} \quad P(a \leq X \leq b) = \int_a^b f(x) dx.$$

Intuition

A distribution tells us not only what values are possible, but how likely each value is.

Examples

→ Suppose

$$p(x) = \begin{cases} k, & x = 0, \\ 2k, & x = 1, \\ 3k, & x = 2, \\ 0, & \text{otherwise.} \end{cases}$$

Find k .

→ Three fair coins are tossed. Let X = number of heads. Define X and find its PMF.

Functions of Random Variables

Often we are interested not in the random variable X itself, but in a function of it.

Examples: $Y = X^2$, $Y = 2X + 3$, $Y = \sqrt{X}$, $Y = \log(X)$,

Suppose X is a random variable and $Y = g(X)$ for some function $g(\cdot)$. Then Y is also a random variable.

Suppose $\frac{X = x}{P(X = x)} \mid \begin{array}{ccc} 0 & 1 & 2 \\ 0.2 & 0.5 & 0.3 \end{array}$. Let $Y = X^2$. Find the distribution of Y .

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Suppose X is a random variable and $Y = g(X)$ for some function $g(\cdot)$. Then Y is also a random variable.

Suppose $\begin{array}{c|ccc} X = x & 0 & 1 & 2 \\ \hline P(X = x) & 0.2 & 0.5 & 0.3 \end{array}$. Let $Y = X^2$. Find the distribution of Y .

Suppose $\begin{array}{c|cccc} X = x & -1 & 0 & 1 & 2 \\ \hline P(X = x) & 0.2 & 0.3 & 0.2 & 0.3 \end{array}$. Let $Y = X^2$. Find the distribution of Y .

Bernoulli Distribution

Models a single experiment with two possible outcomes: Success (1) or Failure (0);

$$X \sim \text{Bernoulli}(p)$$

Probability Mass Function

$$P(X = x) = p^x(1 - p)^{1-x},$$
$$x = 0, 1$$

Parameter: $0 < p < 1$

Mean: $E(X) = p$

Variance: $\text{Var}(X) = p(1 - p)$

Examples

- Coin toss
- Product defective?
- Patient survives?
- Email is spam?

Interpretation

Probability of a single success.

Binomial Distribution

Models the number of successes in n independent Bernoulli trials.

$$X \sim \text{Binomial}(n, p)$$

Probability Mass Function

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x},$$

$$x = 0, 1, \dots, n$$

Parameters: n , $0 < p < 1$

Mean: $E(X) = np$

Variance: $\text{Var}(X) = np(1 - p)$

Examples

- Number of heads in 20 tosses
- Defective items in a sample
- Patients responding to treatment
- Number of successful transmissions

Interpretation

Counts the number of successes in a fixed number of independent trials.

Poisson Distribution

Models the number of events occurring during a fixed interval of time or space.

$$X \sim \text{Poisson}(\lambda)$$

Probability Mass Function

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!},$$

$$x = 0, 1, \dots,$$

Parameter: $\lambda > 0$

Mean: $E(X) = \lambda$

Variance: $\text{Var}(X) = \lambda$

Examples

- Customer arrivals
- Phone calls per hour
- Traffic accidents
- Radioactive decay

Interpretation

Counts independent random events occurring at a constant average rate.

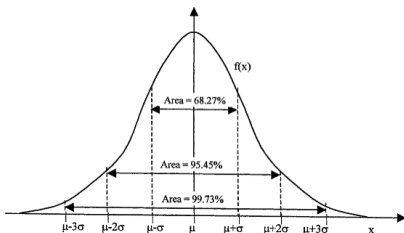
Normal Distribution

The Normal (Gaussian) distribution is the most widely used continuous probability model.

$$X \sim N(\mu, \sigma^2)$$

Probability Density Function

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(x - \mu)^2}{2\sigma^2}\right]$$



Properties

- Parameters: Mean, μ
- Parameters: Variance, σ^2
- Bell-shaped
- Symmetric about μ
- Mean = Median = Mode
- Total area under curve = 1

Role of the Parameters

Changing the Mean

Increasing μ

- shifts the distribution to the right,
- does not change its shape.



Changing the Variance

Increasing σ^2

- increases the spread,
- produces a wider curve.



Standard Normal Distribution

Every normal random variable can be transformed into the **standard normal distribution**.

If $X \sim N(\mu, \sigma^2)$ then $Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$.

The standard normal density is

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}.$$

Why Standardize?

Once transformed to Z , all probabilities can be obtained from a single standard normal table.

Important Facts

Symmetry

$$P(Z < -a) = P(Z > a),$$

$$P(Z < a) = 1 - P(Z > a).$$

Empirical Rule

$$68\% \quad \mu \pm \sigma$$

$$95\% \quad \mu \pm 2\sigma$$

$$99.7\% \quad \mu \pm 3\sigma$$

Useful Values

z	$P(Z < z)$
0	0.5000
1	0.8413
1.96	0.9750
2.576	0.9950

Expectation (Mean)

Expectation summarizes the **centre** or long-run average of a probability distribution.

For a discrete random variable, $E(X) = \sum_x xp(x)$.

For a continuous random variable, $E(X) = \int_{-\infty}^{\infty} xf(x) dx$.

If $Y = g(X)$, $E[g(X)] = \sum_x g(x)p(x)$ (or $\int g(x)f(x) dx$ for contin. RVs).

Important Properties

$$E(aX + b) = aE(X) + b,$$

$$E(X + Y) = E(X) + E(Y).$$

Interpretation

Expectation is the probability-weighted average (or long-run average) of a random variable. It need not be an observable value (e.g., the expected outcome of a die roll is 3.5).

Variance and Standard Deviation

Variance measures the **spread** of a distribution around its mean.

$$\text{Var}(X) = E[(X - E(X))^2] = \text{Var}(X) = E(X^2) - [E(X)]^2.$$

Standard deviation:

$$SD(X) = \sqrt{\text{Var}(X)}.$$

Useful Properties

$$\text{Var}(X + c) = \text{Var}(X), \quad \text{Var}(cX) = c^2 \text{Var}(X).$$

Interpretation

Expectation describes the **location** of a distribution, whereas variance measures its **uncertainty or variability**. Two distributions may have the same mean but very different variances.

Joint and Conditional Distributions

Joint distribution: $P(X = x, Y = y)$.

Marginal distribution: $P(X = x) = \sum_y P(X = x, Y = y)$.

Conditional distribution:

$$P(Y = y \mid X = x) = \frac{P(X = x, Y = y)}{P(X = x)}.$$

Example: Let $X = \text{Rain today}$ and $Y = \text{Heavy traffic}$.

$$P(\text{Rain}, \text{Heavy}) = P(\text{Heavy} \mid \text{Rain}) P(\text{Rain}) \leq \min\{P(\text{Rain}), P(\text{Heavy})\}.$$

Intuition

Conditional distributions describe how uncertainty changes after new information becomes available.

Independence

Definition

Two events A and B are independent if

$$P(A \cap B) = P(A)P(B).$$

Two random variables are independent if

$$f(x, y) = f_X(x)f_Y(y).$$

Example

Independent: Two successive coin tosses?

Important

Independence is a **modeling assumption**, not a property that automatically holds in real data.

Are These Events Independent?

For each pair of events, guess whether they are **Independent** or **Dependent**.

→ Rain today & Heavy traffic.

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→ Two successive tosses of a fair coin.

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Independent

→ Height & Weight of adults.

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→ Ice cream sales & Drowning incidents.

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→ Ice cream sales & Drowning incidents.

Dependent

(Common cause: warm weather)

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→ Two successive tosses of a fair coin.

Independent

→ Height & Weight of adults.

Dependent

→ Ice cream sales & Drowning incidents.

Dependent

(Common cause: warm weather)

→ Shoe size & IQ (entire population).

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→ Two successive tosses of a fair coin.

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→ Height & Weight of adults.

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→ Ice cream sales & Drowning incidents.

Dependent

(Common cause: warm weather)

→ Shoe size & IQ (entire population).

Dependent

(Common cause: age)

Takeaway

Independence is a property of the *joint probability distribution*, not of intuition.

Common Probability Models

Model	Random Variable	Typical Application
Bernoulli	0/1 outcome	Success or failure
Binomial	Number of successes	Clinical trials
Poisson	Count	Customer arrivals
Exponential	Waiting time	Reliability
Normal	Continuous measurement	Heights, measurement errors

Take-home Message

Probability provides a collection of mathematical models for describing uncertainty.

The statistician's first task is to decide which model best represents the data-generating mechanism.